

# QuietKey Gate A Capture Spike, Addendum 01

T0-free reader path and stock OCR baseline, 16 August 2026. Companion to spike-verdict-01; nothing in the verdict is revised, this addendum closes its two named follow-ups.

## 1. T0-free reader path: production geometry anchoring validated

The verdict flagged that the spike selected capture geometry using the known-content T0 control, which does not exist on a production Recovery Document. This run removed every T0 dependency: geometry hypotheses are ranked by a content-free score, and classification uses the global baseline bank only, with no own-sheet exemplar pooling. The score has two parts. The 15 sentinel cells (c, v, 0 at the head of each block, a public protocol constant) act as an alias-rejection gate: a grid shifted by one character pitch reads every window as a genuine neighbouring glyph with high confidence, and only known-content cells detect that. Mean classifier confidence over the five block-head lines, including their line ends, acts as the fine discriminator: sentinel cells all sit at line starts where accumulated pitch error is zero, so sentinel score alone is blind to pitch mistakes; confidence at line ends is not. Sentinel-only selection was tried first and lost three tokens on one sheet to exactly this blindness before the confidence term was added. Result: 116 of 135 tokens decode (92 of 108 treated, 24 of 27 controls, 26 of 27 sheets with at least one position recovered), against 118 of 135 on the T0-selected path. Six tokens flipped:

| sheet              | token | T0-selected | T0-free |
|--------------------|-------|-------------|---------|
| baseline-0-dim-S01 | T2    | decode      | fail    |
| baseline-0-dim-S01 | T3    | decode      | fail    |
| baseline-0-std-S01 | T4    | decode      | fail    |
| coffee-2-std-S04   | T0    | decode      | fail    |
| crumple-1-std-S21  | T4    | fail        | decode  |
| water-3-std-S08    | T1    | fail        | decode  |

Every loss is attributable and none is a geometry-anchor failure. The two dim-capture losses sit on a capture condition the verdict already declared below envelope. The std-S01 T4 loss is the softest baseline capture losing own-T0 pooling help on the bottom of the page. The coffee-2 T0 loss is the token the verdict identified as sitting exactly on the parity budget boundary; it flipped under a marginally different grid. Two tokens were gained. Geometry selection without T0 matched or beat T0 selection on 25 of 27 sheets. Conclusion: the production anchoring design (sentinel gate plus confidence, per-line comb registration) is validated on the existing damage corpus, and the bridge print run is reduced from a risk-retirement exercise to a confirmation exercise. The honest production-representative envelope figure is 116 of 135, not 118. One legitimate recovery option for the pooling loss is noted but deliberately not implemented here: confidence-gated self-pooling of high-confidence cells needs no ground truth and would be a valid terminal behaviour, but it belongs behind its own test coverage.

## 2. Stock OCR baseline: the blueprint 3.2 reader comparison

Tesseract 5.3.4 on the same 27 sheets, deliberately generous conditions: it received the same rectified pages and perfect line localization from the QuietKey geometry, so it is measured on the reading step alone. Variant A is stock single-line mode. Variant B adds a whitelist of the 32 Bech32 characters. Stock OCR has no erasure concept, so every mistake is silent and costs double parity budget; and token assembly requires three exact-length lines, so a single insertion or deletion destroys the positional structure before Reed-Solomon runs. No sentinel repair was applied.

| reader              | char accuracy                         | erasures                | exact-length lines      | tokens decoded |
|---------------------|---------------------------------------|-------------------------|-------------------------|----------------|
| tesseract stock     | 83.3%                                 | none, all errors silent | 192/405                 | 0/135          |
| tesseract whitelist | 86.8%                                 | none, all errors silent | 106/405                 | 2/135          |
| QuietKey T0-free    | about 92% kept explicit, budget-aware |                         | 405/405 by construction | 116/135        |

The whitelist raises character accuracy and worsens structure: forcing every glyph into the charset produces more merges and splits, and exact-length lines drop from 192 to 106 of 405. The two whitelist decodes occurred on near-pristine captures. The comparison closes the section 3.2 requirement at spike scale and states the architectural point plainly: a sequence recognizer is structurally unsuitable for this decoding problem regardless of its accuracy, because positional Reed-Solomon needs a fixed character grid and an erasure channel, and OCR provides neither. This is the empirical justification for the confidence-aware fixed-grid reader that the Gate C audit narrative will need.

## 3. Consequence for the programme

Both named follow-ups from the verdict are closed on existing captures at zero print cost. The remaining Phase 3 sequence is unchanged in content and reduced in risk: bridge print run of boundary treatments on production-format

pages as confirmation, Gate A1 acceptance thresholds frozen before the definitive matrix using the 116 of 135 T0-free envelope as the reference, formal multi-printer multi-camera matrix, Gate A2 salience pilot in parallel. The spike code, per-cell records, and all three result sets should enter the repository and the handoff document should be marked superseded per the single-source-of-truth rule.